The Broadhurst–Roberts Recursion and Bessel Period Pairings

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Abstract

Broadhurst and Roberts gave a recursion for rational matrices appearing in quadratic relations between Bessel moments. Fresán, Sabbah and Yu proved the period relations with a geometric intersection matrix, leaving its identification with that recursion open. We identify the recursion matrix with coefficients of the symmetric-power Bessel operator and then with the inverse de Rham pairing. The argument includes the resonant symmetric powers. In particular, the inverse middle pairing multiplied by $N!$ has integer entries, answering a second question of Fresán–Sabbah–Yu. The main step sums the auxiliary recursion in a period-ratio coordinate; formal residues then recover the intersection form.

1 Introduction and statements

Put $\theta_z = z\,d/dz$ and consider the rank-two connection

$$\theta_z v_0 = v_1, \qquad \theta_z v_1 = z v_0. \tag{1.1}$$

This is the Kloosterman connection Kl_2 in the normalization of Fresán–Sabbah–Yu [3]. For $N \geq 3$, set

$$r = \left\lfloor \frac{N-1}{2} \right\rfloor, \qquad q = r + 1.$$

Let S_N denote their middle de Rham intersection matrix in the basis of [3, Corollary 3.11]. If $4 \nmid N$, its basis is $\omega_i = [z^i v_0^N\, dz/z]$, $1 \leq i \leq r$. If $N = 4\ell$, remove index ℓ and use

$$\omega_i' = \omega_i - \gamma_{N,i}\omega_\ell, \qquad i \in I_N = \{1, \dots, r\} \setminus \{\ell\}, \tag{1.2}$$

where

$$2^N \big(I_0(2\sqrt{z})K_0(2\sqrt{z})\big)^{N/2} \sim \sum_{j \geq \ell} \gamma_{N,j} z^{-j}, \qquad \gamma_{N,\ell} = 1. \tag{1.3}$$

For $4 \nmid N$, let $I_N = \{1, \dots, r\}$. Indices in a submatrix always retain their original order. For $N = 4$ the middle matrix is empty; the identities below use the usual zero-dimensional convention.

Define differential operators in $\mathbb{Z}[z, \theta_z]$ by

$$P_0^- = 1, \quad P_1^- = \theta_z, \qquad P_{k+1}^- = \theta_z P_k^- - k(N+1-k) z P_{k-1}^-, \quad 1 \leq k \leq N. \tag{1.4}$$

Write $\mathcal{L}_N = P^-_{N+1}$ and form the integer matrix

$$E_N(a,b) = [z^a]\mathcal{L}_N z^{-b}, \qquad 1 \le a, b \le r. \tag{1.5}$$

The operator recursion is established: it is the Bronstein–Mulders–Weil construction, as stated and proved in Borwein–Salvy [1, Lemmas 2–3].

Theorem 1.1. *With the above normalization,*

$$N! S_N^{-1} = (-1)^N E_N[I_N, I_N]. \tag{1.6}$$

In particular $N!S_N^{-1}$ has integer entries for every $N \ge 3$.

Broadhurst–Roberts [2, formulas (5.7)–(5.12)] define a rational $r \times r$ matrix D_N^{BR} through a bivariate recursion, recalled in Section 2. Put $T_N = \mathrm{diag}((-4)^1, \dots, (-4)^r)$ and $T_I = T_N[I_N, I_N]$.

Theorem 1.2. *The literal Broadhurst–Roberts recursion satisfies*

$$D_N^{\mathrm{BR}} = T_N^{-1} E_N T_N^{-1}, \tag{1.7}$$

$$D_N^{\mathrm{BR}}[I_N, I_N] = (-1)^N N! \, (T_I S_N T_I)^{-1}. \tag{1.8}$$

Equation (1.8) proves the matrix identifications left open in [3, Remark 5.12]. Equation (1.6) answers the integrality question in the same remark. The quadratic period relations themselves were proved in [3], with a different proof in Zhou [5]. Zhou also distinguishes the recursion question in [5, Section 4.3]. Our argument supplies that identification, using their established geometric pairing and residue formulas. It does not address the conjecture that quadratic relations exhaust all algebraic relations between these periods.

2 Summing the auxiliary recursion

2.1 The local period coordinate

Set $x = t^2/4$ and

$$J(x) = J_0(2\sqrt{x}) = \sum_{n \ge 0} \frac{(-1)^n x^n}{(n!)^2}, \qquad \theta = x\frac{d}{dx}.$$

Thus $(\theta^2 + x)J = 0$. Define formal series

$$C(x) = \frac{2(1 - J^2 + 2xJJ')}{3x^2}, \qquad V = \frac{J^2}{C} = \sum_{n \ge 0} \frac{v_n x^n}{n!}, \qquad W = -\frac{JJ'}{C} = \sum_{n \ge 0} \frac{w_n x^n}{n!}. \tag{2.1}$$

They have constant term one. These are exactly the series in [2, formulas (5.7)–(5.9)]: the change of variable uses $J_0'(t) = -J_1(t)$.

There is a unique $\beta \in x\mathbb{Q}[[x]]$ such that

$$\beta' = \frac{J^{-2} - 1}{x}. \tag{2.2}$$

Put $w = \log x + \beta$. The two solutions J, Jw have θ-Wronskian one, since $\theta w = J^{-2}$. In particular,

$$Q = x \exp \beta, \qquad \frac{dQ}{Q} = \frac{dx}{xJ^2}, \qquad Q\frac{d}{dQ} = J^2\theta. \tag{2.3}$$

Here $Q = x + O(x^2)$ has a formal compositional inverse.

2.2 The bivariate polynomials

The polynomials of [2, formula (5.10)] have $H_0 = 1$, $H_{-1} = 0$, and, for $s \geq 1$,

$$H_s(Y,Z) = ZH_{s-1} - (s-1)YH_{s-2} - \sum_{k=1}^{s-1} \binom{s-1}{k} \big(v_k H_{s-k} - w_k Z H_{s-k-1}\big). \tag{2.4}$$

Set

$$d_s(N,c) = \frac{H_s(3c/2, N+2-2c)}{4^s s!}, \qquad D_N^{\mathrm{BR}}(a,b) = \sum_{c=-b}^{a} d_{a-c}(N,-c) d_{b+c}(N,c) c^{N+1}. \tag{2.5}$$

Lemma 2.1. *For every integer c,*

$$\sum_{s \geq 0} d_s(N,c)(4x)^s = J^{-N-2} \exp(-c\beta). \tag{2.6}$$

Proof. Let $H(x) = \sum_s H_s(Y,Z) x^s/s!$. Multiply (2.4) by $x^{s-1}/(s-1)!$ and sum. The convolution terms are $(V-1)H'$ and $Z(W-1)H$, giving

$$VH' = (ZW - Yx)H.$$

From (2.1) and (2.2),

$$\frac{W}{V} = -\frac{J'}{J}, \qquad \frac{xC}{J^2} = \frac{2}{3}\left(\beta' + 2\frac{J'}{J}\right).$$

For $Y = 3c/2$ and $Z = N+2-2c$, it follows that $H'/H = -(N+2)J'/J - c\beta'$. The initial value is $H(0) = 1$, which proves (2.6). □

Proposition 2.2. *Define*

$$L_N = J^{-N-2}(J^2\theta)^{N+1} J^{-N}. \tag{2.7}$$

Then, for all $a, b \geq 1$,

$$D_N^{\mathrm{BR}}(a,b) = 4^{-a-b}[x^a] L_N x^{-b}. \tag{2.8}$$

Proof. Put $f = J^{-N-2}$. Lemma 2.1 rewrites $4^{a+b} D_N^{\mathrm{BR}}(a,b)$ as the coefficient of x^a in

$$f(x) \sum_{c \in \mathbb{Z}} c^{N+1} Q(x)^c g_c, \qquad g_c = [y^{b+c}] f(y) \exp(-c\beta(y)).$$

Only $-b \leq c \leq a$ contributes. In the residue defining g_c, make the formal substitution $Q = y \exp \beta(y)$. Equation (2.3) gives

$$g_c = \mathrm{res}_{y=0}\, f(y) y^{-b} Q^{-c} \frac{dy}{y} = [Q^c] y(Q)^{-b} J(y(Q))^{-N}.$$

Multiplication of the cth coefficient by c^{N+1} is the operator $(Q\, d/dQ)^{N+1}$. Substituting back and using (2.3) gives exactly (2.7). All operations concern formal Laurent coefficients; no summability assumption is involved. □

3 The symmetric-power operator

Lemma 3.1. *The operator L_N is the monic symmetric power of $\theta^2 + x$. It belongs to $\mathbb{Z}[x, \theta]$, has x-degree at most q, and*

$$L_N^* = (-1)^{N+1} L_N \tag{3.1}$$

for the formal adjoint associated with dx/x. Moreover $\mathcal{L}_N = L_N|_{x=-z}$.

Proof. For completeness, apply the established recursion to $u_k = y^{N-k}(\theta y)^k$, where $(\theta^2 + x)y = 0$. Since $\theta u_k = (N-k)u_{k+1} - kxu_{k-1}$, the operators

$$P_0 = 1, \quad P_1 = \theta, \qquad P_{k+1} = \theta P_k + k(N+1-k)xP_{k-1}$$

satisfy $P_k u_0 = N! u_k/(N-k)!$ for $k \le N$ and $P_{N+1}u_0 = 0$. Varying y through linear combinations of J and Jw shows that P_{N+1} kills their full symmetric power. The same is true of (2.7): multiplication by J^{-N} gives a polynomial of degree at most N in w, and $J^2\theta w = 1$. Both operators are monic of order $N+1$. Their common solution space has dimension $N+1$, so they coincide. This proves the integer-coefficient assertion and the substitution claim. Induction in the recursion bounds the weight of a term $x^j\theta^k$ by $2j + k \le N + 1$.

To compute the adjoint, put $A = J^2\theta$. Since $\theta^* = -\theta$, $A^* = -J^{-2}AJ^2$. Reverse the factors in (2.7) and cancel the intervening powers of J. This gives (3.1). □

The monic operator is independent of the chosen solution frame: every change of frame preserves the symmetric-power solution space. Thus the expression in local periods (2.7) descends to the integer operator (1.4). The local series in the recursion determine coefficients of this common invariant. This is the useful period-frame step of the proof.

Lemma 3.2. *Write $\mathcal{L}_N = \sum_{j=0}^{q} z^j p_j(\theta_z)$. Then*

$$p_q(\theta_z) = \begin{cases} (-1)^q((2q-1)!!)^2, & N = 2q-1, \\ (-1)^q(2^q q!)^2(\theta_z + q/2), & N = 2q. \end{cases} \tag{3.2}$$

The matrix E_N satisfies

$$E_N^T = (-1)^{N+1} E_N, \qquad E_N(a,b) = 0 \quad (a+b > q).$$

It is invertible if $4 \nmid N$. If $N = 4\ell$, it has rank $r-1$, and its principal submatrix on I_N is invertible.

Proof. For $N = 2q-1$, the top coefficient of z^j in P_{2j}^- is $(-1)^j \prod_{h=1}^{j}(2h-1)(N+2-2h)$, directly from the recursion. This gives the first formula.

For $N = 2q$, the weight bound gives $\deg p_q \le 1$. In the associated commutative symbol, derivative commutators have lower weight. The recurrence becomes the determinant recurrence for the tridiagonal symmetric-power matrix with off-diagonal entries $N-a$ and az. Its eigenvalues are $(N-2a)\sqrt{z}$, $0 \le a \le N$: these are the eigenvalues of the symmetric power of the matrix with eigenvalues $\sqrt{z}, -\sqrt{z}$. The coefficient of $z^q\theta_z$ in its characteristic polynomial is $(-1)^q(2^q q!)^2$. Adjointness requires $p_q(-\theta_z - q) = -p_q(\theta_z)$, fixing its constant term.

The adjoint identity, applied to the constant term of $z^{-a}\mathcal{L}_N z^{-b}$, proves the stated symmetry of E_N. The degree bound gives its zero entries. On the anti-diagonal $a + b = q$ its entries are $p_q(-b)$. They are nonzero for odd N and for $N \equiv 2 \pmod 4$. When $N = 4\ell$, only the central anti-diagonal entry vanishes; deleting its index leaves an invertible anti-triangular matrix. The full matrix is skew-symmetric of odd size, so its rank is exactly $r-1$. □

Proposition 2.2 and the substitution $x = -z$ now give

$$E_N(a,b) = (-4)^{a+b} D_N^{\mathrm{BR}}(a,b). \tag{3.3}$$

This proves (1.7).

4 Formal primitives and the intersection matrix

We use the existence of formal primitives and the residue description of the pairing from [3, Lemma 3.3, Proposition 3.8 and Section 3.b]. We recall the normalization to keep track of the sign and transpose. For $u_a = v_0^{N-a} v_1^a$, their algebraic pairing is

$$\langle u_a, u_b \rangle = \begin{cases} (-1)^a / \binom{N}{a}, & a+b = N, \\ 0, & a+b \neq N. \end{cases} \tag{4.1}$$

The primitive at zero has coefficients in $z\mathbb{C}[[z]]$ for the positive-power sources used here, so its residue contribution is zero.

Suppose a formal primitive at infinity is $m = \sum_{a=0}^{N} f_a u_a$ and $\nabla m = g(z) u_0 \, dz/z$. Its components satisfy

$$\begin{aligned} \theta_z f_a + (N-a+1) f_{a-1} + (a+1) z f_{a+1} = 0 \qquad & (1 \le a \le N), \\ \theta_z f_0 + z f_1 = g. & \end{aligned} \tag{4.2}$$

Put $\eta = f_N$.

Lemma 4.1. *These equations imply*

$$\mathcal{L}_N \eta = (-1)^N N! g. \tag{4.3}$$

If $\deg g \le q-1$*, every formal Laurent solution* η *at infinity has only strictly negative powers of* z*.*

Proof. The equations with $a \ge 1$ determine the other components from η. The coefficient of $\theta_z^{N+1}\eta$ in the last equation is $(-1)^N/N!$. The homogeneous eliminant has as solutions the top components of horizontal symmetric-power sections. For a horizontal rank-two section, its top component solves $(\theta_z^2 - z)y = 0$. Products of those sections therefore give exactly the full symmetric-power solution space in the top component. The monic eliminant is consequently $\mathcal{L}_N$, proving (4.3) including its factor.

Write the highest exponent of a nonzero Laurent series η as M. If $M \ge 0$, Lemma 3.2 gives $p_q(M) \neq 0$. The highest exponent on the left of (4.3) is then $M + q > \deg g$, a contradiction. □

Let $M_{bi} = [z^{-b}]\eta_i$ for a family of primitives with sources g_i. A term z^{-b} with $b \ge q$ cannot contribute to a positive power under $\mathcal{L}_N$. The lemma therefore gives the finite identity

$$E_N M = (-1)^N N! G, \tag{4.4}$$

where $G_{ai} = [z^a] g_i$, $1 \le a \le r$.

4.1 Nonresonant powers

Suppose $4 \nmid N$ and take $g_i = z^i$. Then $G = 1$ and $M = (-1)^N N! E_N^{-1}$. By (4.1), the residue at infinity is

$$S_N(i,j) = (-1)^{N+1} M_{ji}; \tag{4.5}$$

this is [3, formula (3.15)]. Thus

$$S_N = -N! E_N^{-T}, \qquad N! S_N^{-1} = (-1)^N E_N,$$

where the second equality uses Lemma 3.2.

4.2 Resonant powers

Suppose $N = 4\ell$ and let $h = (\gamma_{N,1}, \dots, \gamma_{N,r})^T$. The formal series in (1.3) solves $\mathcal{L}_N h(z) = 0$: it is the product of $N/2$ copies of each scalar Bessel solution. Taking positive coefficients gives $E_N h = 0$. Since $h_\ell = 1$, Lemma 3.2 shows that this vector spans the kernel.

Let U be the $r \times (r-1)$ matrix whose columns are

$$U_i = e_i - h_i e_\ell, \qquad i \in I_N. \tag{4.6}$$

For the middle sources $g_i = z^i - h_i z^\ell$, equation (4.4) becomes $E_N M = N! U$. The ambiguity of a primitive is a multiple of the horizontal formal section whose top component is (1.3); see [3, Proposition 3.8]. We may therefore impose the gauge $M_{\ell i} = 0$. Writing $\bar{E} = E_N[I_N, I_N]$ gives

$$M[I_N, I_N] = N! \bar{E}^{-1}.$$

The residue with the corrected source is

$$S_N(i,j) = -(M_{ji} - h_j M_{\ell i}) = -M_{ji}.$$

Adding a horizontal section does not change it, since $h_j - h_j h_\ell = 0$. Thus the gauge computes the same pairing as [3, formula (3.16)], independently of their normalization of the primitive. We obtain

$$S_N = -N! \bar{E}^{-T} = N! \bar{E}^{-1}, \qquad N! S_N^{-1} = \bar{E}.$$

This proves Theorem 1.1 in all cases. Combining it with (3.3) proves (1.8) and Theorem 1.2.

5 The quadratic relation, including the singular matrix

For clarity we also recover the original normalization. Let $M(a,b,c) = \int_0^\infty I_0(t)^a K_0(t)^b t^c \, dt$ and $Z(s) = |B_s|/s!$, with B_s the Bernoulli number. The matrices in [2, formulas (5.2)–(5.6)] are

$$\begin{aligned}
F_{2r+1}(u,a) &= \frac{(-1)^{a-1}}{\pi^u} M(r+1-u, r+u, 2a-1), \\
F_{2r+2}(u,a) &= \frac{(-1)^{a-1}}{\pi^{u+1/2}} M(r+1-u, r+1+u, 2a-1), \\
B^{\mathrm{BR}}_{2r+1}(u,v) &= (-1)^{u+r} 2^{-2r-2} (r+u)!(r+v)! Z(u+v), \\
B^{\mathrm{BR}}_{2r+2}(u,v) &= (-1)^{u+r} 2^{-2r-3} (r+u+1)!(r+v+1)! Z(u+v+1),
\end{aligned}$$

with all indices from 1 to r.

Corollary 5.1. *For every $N \geq 3$, with the literal recursion matrix,*

$$F_N D_N^{\mathrm{BR}} F_N^T = B_N^{\mathrm{BR}}.$$

Proof. If $4 \nmid N$, this follows at once from Theorem 1.2 and [3, Corollary 5.10]. We give the pullback argument needed for $N = 4\ell$. Write P and B for the $r \times r$ matrices of [3, Sections 4–5], retaining indices $1, \dots, r$. The period matrix on the middle sources, with all these homology rows, is $Q = PU$. Both its rows and the pairing B factor through the map from rapid-decay homology to middle homology: $\alpha_i \mapsto \beta_i$, as in [3, formula (4.5)]. The basis of the latter space is $\beta_2, \dots, \beta_r$. If A records the coordinates of all β_i in that basis, then

$$Q = AP^{\mathrm{mid}}, \qquad B = AB^{\mathrm{mid}}A^T.$$

The established middle period relation [3, Theorem 5.3] consequently gives

$$QS_N^{-1}Q^T = (-2\pi\mathrm{i})^{N+1}B.$$

The identity $E_N h = 0$, with $h_\ell = 1$, implies $E_N = U\bar{E}U^T$: the blocks on $I_N \times I_N$ agree, and the kernel determines the remaining row and column. Since $S_N^{-1} = \bar{E}/N!$, we obtain

$$PE_N P^T = N!(-2\pi\mathrm{i})^{N+1}B.$$

Finally the normalizations in [3, formula (5.9)] are

$$F_N = (-2\sqrt{\pi})^{-N-1}R^T P T_N, \qquad B_N^{\mathrm{BR}} = -\frac{\mathrm{i}^{N+1}N!}{2^{N+1}}R^T B R,$$

where $R = \mathrm{antidiag}(\mathrm{i}, \dots, \mathrm{i}^r)$. Substitute (1.7). The scalar factor is $(-2\sqrt{\pi})^{-2N-2}N!(-2\pi\mathrm{i})^{N+1} = -\mathrm{i}^{N+1}N!/2^{N+1}$ because N is even. This proves the full singular relation as well. □

6 Examples and the role of the period frame

For $N = 5, 6$, the formulas give

$$D_5^{\mathrm{BR}} = \begin{pmatrix} 13/8 & 225/64 \\ 225/64 & 0 \end{pmatrix}, \quad S_5 = \begin{pmatrix} 0 & 8/15 \\ 8/15 & 208/3375 \end{pmatrix},$$

$$D_6^{\mathrm{BR}} = \begin{pmatrix} 0 & -18 \\ 18 & 0 \end{pmatrix}, \qquad S_6 = \begin{pmatrix} 0 & -5/8 \\ 5/8 & 0 \end{pmatrix}.$$

The pairings reproduce [3, Examples 3.19–3.20]. For $N = 8$, the first nontrivial resonant example is

$$D_8^{\mathrm{BR}} = \begin{pmatrix} 0 & -288 & -576 \\ 288 & 0 & 0 \\ 576 & 0 & 0 \end{pmatrix}, \qquad S_8 = \begin{pmatrix} 0 & 35/128 \\ -35/128 & 0 \end{pmatrix},$$

with $I_8 = \{1, 3\}$ and $h = (0, 1, 1/8)^T$. The omitted row and column reflect a horizontal formal solution, not a failure of the coefficient formula.

The proof separates a local series presentation from the invariant it determines. A determinant-one frame (J, Jw) belongs to the torsor of normalized solution frames of the rank-two connection. The period ratio supplies Q, in which the convolution in (2.5) becomes an Euler derivative. Returning to x gives (2.7); frame invariance then identifies it with the integer symmetric-power operator. The

established residue pairing turns its finite coefficients into S_N^{-1}. Thus the frame comparison connects the local recursion with a global intersection form, identifying the Broadhurst–Roberts recursion and proving the factorial-scaled integrality in Theorems 1.1–1.2.

This is a separate application of the period-frame approach used for quartic periods in [4]; here it is applied to the symmetric-power Bessel connection.

The accompanying exact checker compares the literal recursion with (1.4) for $3 \le N \le 32$. An independent downward elimination in the full connection checks the residue signs and normalization, including the corrected sources and kernel in every resonant case in that range. All factorial-scaled inverse matrices are integral, as the theorem requires. These finite checks supplement the formal proofs above. The source and data are in `research/bessel-recursion`; `make bessel-recursion-check` reproduces them.

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